

Lesson 3

Programming Sensors



Input, decisions, then real sensor data.

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- 01 Quick review of Lesson 2
 - 02 Example: mini calculator
 - 03 Classwork: larger number & grades
 - 04 Sensors: reading & cleaning data
 - 05 Counting button presses
-

Input, process, output.

Input

The data that we enter or the computer gets from the environment via sensors.

FROM THE USER

```
while(Serial.available()==0){}  
int a = Serial.parseInt();
```

Wait when there's no input, then read it and store it in "a".

Process

Our codes (orders) for the computer to process.

Output

The response from the computer after processing the codes.

FROM A SENSOR

```
incipe.main();  
incipe.getTemperature();
```

Activate the sensors & sensor libraries, then read a value.

Compare, combine, decide.

COMPARE

Symbol	Meaning
<code>==</code>	Equals to
<code>!=</code>	Not equals to
<code><=</code>	Smaller than or equal to
<code>>=</code>	Larger than or equal to
<code><</code>	Smaller than
<code>></code>	Larger than

Remember: `=` assigns a value; `==` compares two values.

COMBINE

`&&` AND: both must be true

`||` OR: any one is true

`!` NOT

DECIDE

```
if (...) { } else if (...) { } else { }
```

Three questions before you code.

- 1 How to ask users to input two numbers?
How to store them?

```
Serial.println("Please type your first number.");  
while(Serial.available()==0){}  
int a = Serial.parseInt();
```

- 2 How to add the two numbers? Store it
to where?

```
int sum = a+b;
```

- 3 How to show the answer?

```
Serial.print("The answer is:");  
Serial.println(sum);
```

The full sketch.

```
void setup() {  
  Serial.begin(9600);  
}  
  
void loop() {  
  Serial.println("Please type your first number.");  
  while(Serial.available()==0){}  
  int a = Serial.parseInt();  
  Serial.println(a);  
  Serial.println("Please type your second number.");  
  while(Serial.available()==0){}  
  int b = Serial.parseInt();  
  Serial.println(b);  
  int sum = a+b;  
  Serial.print("The answer is:");  
  Serial.println(sum);  
}
```

INPUT

Two numbers from the serial monitor.

PROCESS

Add them and store the result in sum.

OUTPUT

Print the answer.

Which number is larger?

```
void loop() {  
  Serial.println("Please enter your first number.");  
  while(Serial.available()==0){}  
  int a = Serial.parseInt();  
  Serial.println(a);  
  Serial.println("Please enter your second number.");  
  while(Serial.available()==0){}  
  int b = Serial.parseInt();  
  Serial.println(b);  
  if(a>b){  
    Serial.print(a);  
    Serial.println(" is the larger number");  
  }  
  else{  
    Serial.print(b);  
    Serial.println(" is the larger number");  
  }  
}
```

YOUR NOTES

Refer to your note page 2.

THINK ABOUT IT

What prints when both numbers are the same?

Turn a score into a grade.

```
void loop() {  
  Serial.println("Please enter your score.");  
  while(Serial.available()==0){}  
  int score = Serial.parseInt();  
  Serial.println(score);  
  
  if(score ≥ 85 && score ≤ 100){  
    Serial.println("Your grade is A");  
  } else if(score ≥ 75 && score ≤ 84){  
    Serial.println("Your grade is B");  
  } else if(score ≥ 60 && score ≤ 74){  
    Serial.println("Your grade is C");  
  } else if(score < 60){  
    Serial.println("Your grade is D");  
  } else {  
    Serial.println("The full mark should be 100, please enter again.");  
  }  
}
```

GRADES

A 85 – 100

B 75 – 84

C 60 – 74

D Below 60

Anything above 100 falls to else.

One function per sensor.

WHAT A SENSOR DOES

A sensor detects and measures changes in our environment and converts them into electrical signals, which can be used to provide information, control the outputs, or trigger some useful action.

Sensor	Function
Ultrasonic	<code>incipe.getDistance()</code> · cm
Temperature	<code>incipe.getTemperature()</code>
Humidity	<code>incipe.getHumidity()</code>
Light	<code>incipe.getLightIntensity()</code>
Air quality	<code>incipe.getPPM()</code>
Alcohol	<code>incipe.getAlcoholPPM()</code>
CO gas	<code>incipe.getCOPPM()</code>
Flame	<code>incipe.getFlameIntensity()</code>
Button	<code>incipe.getButtonResponse()</code> · 1 = pressed

Turn raw numbers into meaning.

Data cleaning is the process of fixing or removing incorrect or incomplete raw data. Here, we focus on how to interpret the raw data to make it more useful.

```
incipe.getLightIntensity() ➡ 700, 1000...
```

A raw reading means little on its own. if-else gives it a label people understand.

AIR QUALITY, IN PLAIN STEPS

```
if AQvalue < xxxx  
    print Good  
else if AQvalue > xxxx and < xxxx  
    print Moderate  
else  
    print Poor
```

Students choose the xxxx boundaries.

At 30°C, what is shown at "temp1"?

READ THE CODE

```
float temp = incipe.getTemperature();  
if (temp>30){  
    incipe.onscreen("temp1", "hot");}  
else if (temp<10){  
    incipe.onscreen("temp1", "cold");}  
else {  
    incipe.onscreen("temp1", temp);}
```

ANSWER

30

WHY

30 is not larger than 30, and not smaller than 10.
So else runs and shows the temperature itself.

The light reading jumps from 100 to 1000. What shows at "light1"?

READ THE CODE

```
float light = incipe.getLightIntensity();  
if (light ≥ 700){  
    incipe.onscreen("light1", "bright");}  
else if (light ≥ 300 && light < 700){  
    incipe.onscreen("light1", "normal");}  
else {  
    incipe.onscreen("light1", "dark");}
```

ANSWER

bright

WHY

1000 is larger than or equal to 700, so the first condition is true.

Count each press once.

FIRST TRY

```
int a = 0;
if (incipe.getButtonResponse() == 1)
{
    a=a+1;
    Serial.println(a);
}
```

1 = true (pressed). loop() runs many times while you hold the button, so a keeps going up.

COUNT ON RELEASE

```
int a = 0;
bool detect = false; // set as false at beginning
if (incipe.getButtonResponse() == 1) // pressed
{
    detect = true;
}
if (incipe.getButtonResponse() == 0 && detect == true)
{ // not pressed and detect is true
    a=a+1;
    Serial.println(a);
    detect = false;
}
```

detect remembers the press; a goes up once, when you let go.

Which sensor haven't we introduced yet?

COVERED SO FAR

Ultrasonic · Temperature & humidity · Light · Air quality · Alcohol
· CO gas · Flame · Button

ANSWER

Sound sensor

WHY

It is the one sensor in the set not covered yet. It comes next.

Four ideas we will learn next.

The sound sensor uses `incipe.getSoundIntensity()`. Its example sketch samples sound for a short time, stores the counts and averages them into Loud, Normal or Quiet.

Timer	<code>PreviousTime = millis();</code>
While loop	<code>while (millis() - PreviousTime < SAMPLE_TIME) { ... }</code>
Array	<code>float database[100];</code>
For loop	<code>for (int i = 1; i < 100; i++) { ... }</code>

Three things to take away.

- 1 Wait, read, store: `while(Serial.available() > 0){}` then `Serial.parseInt()`.
- 2 `if`, `else if` & `else` with `&&` turn a number into a decision, like a grade.
- 3 Sensor readings are just numbers: the same `if-else` turns them into labels.

Thank you!